



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : MPBA G526
Course Title : Pricing Analytics
Instructor-in-Charge : Praveen Goyal
Instructor(s) :
Tutorial/Practical Instructors:

1. Course Description:

This course provides an introduction to both the theory and the practice of revenue management and pricing, Fundamentals of price theory, Segmentation and price differentiation, Break- even analysis, Price sensitivity and willingness-to-pay, Empirical estimations of price-response functions, Price optimization, Markdown optimization, The hedonic pricing model, Revenue Management, Monte Carlo simulation for pricing decisions, Conjoint analysis for pricing decisions.

2. Scope and Objective of the Course: The objectives of this course are: (1) To make you acquainted with the concept of pricing and how data can help in setting the right price for the organization. This will be done primarily through lectures, class discussions and assigned readings.

3. Text Books:

T1: Pricing analytics, Model and advanced quantitative techniques for product pricing by Walter R. Paczkowski, Routledge

T2: Pricing and revenue optimization by Robert L. Phillips, Stanford Business Books

4. Reference Books:

R1: Pricing segmentation and analytics by Tudor Bodea and Mark Ferguson, Routledge

5. Course Plan:

Lecture No.	Learning Objectives	Topics to be covered	Reference
1-3	Introduction and overview	Introduction to pricing, fundamentals of theory of pricing and the practice of revenue management and pricing, impact of price on profit, Pricing analytics, Qualitative and quantitative pricing, Pricing analytics process, role of data in pricing, customer behavior aspect of pricing	Chapter 1 in T1 and T2
4-5	Role of elasticity in pricing	Elasticity concept, different types of elasticity, consumer surplus, role of elasticity in pricing, cost in pricing	T1: Chapter 2 and Chapter 3
6-8	Pricing for different market segments	Price differentiation, Degrees of price differentiation, economic theory of price differentiation, Segmentation and price differentiation	T1: Chapter 8
9-11	Break- even analysis, and willingness-to-pay	Understanding of break-even analysis, break-even analysis of price changes, break-even with cost changes, Willingness to pay, price response function	Class Notes



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12-15	Empirical estimations of price-response functions	Sources of price and demand data, empirical estimation of the price response function, Constant price elasticity function	T2: Chapter 3
16-22	Price optimization, Markdown optimization	Basic price optimization, price optimization with capacity constraints, Time based price differentiation, Elasticity and optimization, Markdown Optimization,	T2: Chapter 3
23-26	The hedonic pricing model	Price hedonism, application of hedonic pricing	Class Notes
27-34	Revenue Management	Importance of revenue management, Levels of revenue management, revenue management strategy, two class model, N-class model, Nested booking, Expected Net contribution, Network management, Incidence Variable	T2: Chapter 6, Class Notes
35-39	Conjoint analysis for pricing decisions	Price response function estimation with conjoint analysis	T1: Chapter 4
40-42	Legal and ethical issues	Price presentation, hiding the price and bounded rationality, asymmetric evaluation, ethical issues in pricing	Class Notes

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	20	06/10 FN2	Close Book
Comprehensive Examination	3 h	40	05/12 AN	Close Book
Assignment/Project/ Case Studies		40		Open Book

Chamber Consultation hour: Will be informed in the class later

Notices: All the notices concerning the course will be communicated on institute email id.

Academic Honesty and Integrity Policy: Academic honesty and integrity are to be maintained by all the students throughout the semester and no type of academic dishonesty is acceptable.

INSTRUCTOR-IN-CHARGE